

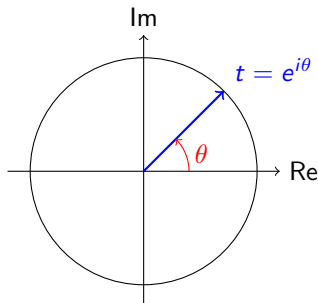
A Direct Proof of Morley's Theorem

An Algebraic Approach Using Complex Numbers

Based on H. Martyn Cundy's Paper

The Foundation: "Turns" in the Complex Plane

- Instead of traditional geometry, this proof maps the triangle onto the complex number plane (the Argand diagram).
- It relies on the concept of a **turn**: a complex number with a unit modulus, representing a point on the unit circle U .
- If a turn is defined as $t = \cos \theta + i \sin \theta$, it algebraically represents a rotation about the origin through angle θ .
- Using turns allows us to manipulate rotations using simple multiplication rather than messy trigonometric angle addition.



The Algebraic Test for Equilateral Triangles

- To test if three points form an equilateral triangle, we use the specific turn ω .
- Let $\omega = \cos(2\pi/3) + i \sin(2\pi/3)$, which corresponds to a 120° rotation.
- Note that $\omega^3 = 1$ and $1 + \omega + \omega^2 = 0$.

The Equilateral Condition

Points with complex affixes p, q , and r form an equilateral triangle if and only if they satisfy:

$$p + \omega q + \omega^2 r = 0 \quad (\text{anticlockwise orientation})$$

or

$$p + \omega^2 q + \omega r = 0 \quad (\text{clockwise orientation})$$

Parameterizing the Triangle and Trisectors

- The circumcircle of triangle ABC is set as the unit circle U .
- To avoid fractional powers later, the vertices A, B , and C are assigned the affixes a^3, b^3 , and c^3 .
- A rotation from A to B is associated with the turn b/a . Therefore, exactly one-third of that rotation is simply the cube root.
- By carefully applying the cube roots of unity (ω), the specific turns corresponding to the exact trisection points on the unit circle can be listed (e.g., a^2b, ab^2, b^2c).

The Grand Conclusion

- Using the standard equation for a chord in the complex plane, $z + tuz^* = t + u$, we find the intersection points of the adjacent trisectors.
- Solving these linear equations gives the affixes for the three intersection points, denoted as x, y , and z .
- Finally, substitute these massive algebraic expressions for x, y , and z into our equilateral test:

$$x + \omega y + \omega^2 z$$

- Because the points originated from a closed triangle, the polynomial factors perfectly and collapses entirely to 0.
- **Conclusion:** $x + \omega y + \omega^2 z = 0$, proving that the Morley triangle xyz is strictly equilateral.